

Kaitlyn Baker

(484) 643-9169 | kaitlynbaker@gmail.com | kmb7652@psu.edu

I am a highly motivated research biologist with experience in designing and conducting field research, leading teams in collection and processing of biological samples from a range of vertebrate hosts, and managing laboratory operations and data integrity. I am passionate about pursuing graduate studies and a career in the intersection of my interests: neurobiology, disease ecology, and field cognition.

EDUCATION

The Pennsylvania State University
B.S., Biology, Eberly College of Science

2024

PROFESSIONAL EXPERIENCE & RESEARCH

Research Technologist – Department of Biology, The Pennsylvania State University *November 2024 – Present*

- Led and coordinated field teams in the collection of biological samples from wild animal populations across multiple field sites spanning five states
- Trained and supervised undergraduate research assistants in advanced field techniques, safety, and data management
- Streamlined logistics for complex, remote fieldwork, collaborating with veterinarians and wildlife professionals to ensure project success
- Managed laboratory operations, including inventory, procurement, and maintenance of equipment

IRES Manifestations of Climate Change – Peru

June 2024 – July 2024

- Investigated ecological impacts of climate change on plant-pollinator interactions across diverse Peruvian ecosystems
- Studied in situ movement patterns of foraging pollinators and the latitudinal variation of pollinator thermal tolerance
- Collected, identified, and curated entomological specimens; performed systematic environmental data analysis

African Savanna Biology – Kenya

December 2023 – January 2024

- Studied the ecology of species interactions, adaptations, and selective pressures that shape community structure in the Maasai Mara National Reserve in southern Kenya
- Led behavioral observations and ecological data collection in challenging field conditions
- Enhanced science outreach through photography and documentary film

Research Assistant – Department of Entomology, The Pennsylvania State University

May 2023 – October 2023

- Studied relationships among plants, insects, and natural enemies to understand factors that regulate insect populations
- Implemented systems to learn about sustainable tactics for pest management and emerging contaminants in aquatic systems
- Conducted plant and arthropod identification, greenhouse experiments, and fly husbandry

TRAININGS & CERTIFICATIONS

- IACUC Training - CITI wildlife no surgery (for non-reptile users)
- IACUC Training - CITI researchers (NOT involved in survival surgery)
- OSHA Bloodborne Pathogens
- EHS Training - CITI biosafety
- Occupational Medicine Assessment
- Retro-orbital sinus blood collection and terminal eye bleeding

TECHNICAL & FIELD SKILLS

- Advanced wildlife handling and sampling (nasal, oral, anal swabs)
- Collection of blood samples via retro-orbital sinus bleeding and jugular withdraws from live animals
- Field team leadership and training
- RNA extraction and storage
- Organization and maintenance of large quantities of samples
- Data analysis, manipulation, and proficiency in Microsoft Office suite

PUBLICATIONS

- Baker, K., Plowright, R.A., Friant S., Hudson P. (In prep) Does the source of a spillover outbreak matter? Comparison of Bats and Rodents zoonotic infections. Will submit to Nature Microbiology Reviews
- Hudson, P.J. & K. Baker 2025. The Nile monitor: Africa's aquatic giant. PT Aware. August 2025.

EXTRACURRICULAR

National Society for Leadership and Success (NSLS)

Fall 2020 – Spring 2024

- Actively participated in Success Networking Teams, supporting peers in academic and personal goal achievement

Network for Excellence in Undergraduate Science (NEXUS) – Vice President

Fall 2021 – Fall 2023

- Fostered academic and professional networking for science undergraduates
- Organized guest lectures and career panels; planned major events for the Eberly College of Science
- Led biweekly meetings and study hours for members to focus on schoolwork

Lion's Pantry Volunteer

Spring 2022 – Fall 2023

- Supported campus food security initiatives through direct service and outreach